Skills Summary

Transformations, Compositions and Inverses Together

Use this reference while completing your worksheet or when studying.

One idea, three faces. A transformation is f composed with a simple linear map; an inverse is the composition that undoes one.

Inside changes x, and lies about direction: $f(x-h)$ moves right by h; $f(kx)$ compresses horizontally by $1/k$; $f(-x)$ reflects in the y-axis.

Outside changes y, and means what it says: $af(x)+c$ stretches vertically by a and shifts up by c; $-f(x)$ reflects in the x-axis.

Order matters when a stretch and a vertical shift meet: stretch first, then shift.

1. Transform a rule

Substitute the inside change for every x in the rule, then apply the outside changes to the whole result. Read the direction from the position, never from the sign alone: $f(x+1)$ moves *left* 1.

Example: $f(x) = x^2$. Write $g(x) = f(x+1) - 4$ as an expanded polynomial.
Step 1. The $+1$ sits inside f, so it is a horizontal move (left 1), and the -4 sits outside, so it lowers the whole graph by 4.
Step 2. Substituting: $g(x) = (x+1)^2 - 4 = x^2 + 2x + 1 - 4$.
Collecting, $g(x) = \boldsymbol{x^2 + 2x - 3}$ — the parabola moved left 1 and down 4.

Try it: $f(x) = x^2$. Write $h(x) = 2f(x-3)$ as an expanded polynomial.

check: $\boldsymbol{2x^2 - 12x + 18}$

2. Compose two functions

$(f \circ g)(x) = f(g(x))$: evaluate the *inner* function first, then feed its whole output into the outer one. The order is not interchangeable — $f \circ g$ and $g \circ f$ are usually different functions. Any input the inner function rejects, or that produces a value the outer function rejects, is outside the domain of the composition.

Example: $f(x) = 3x - 1$ and $g(x) = x^2 + 2$. Find $(f \circ g)(2)$.
Step 1. Composition runs inside out, so g must be evaluated before f ever sees the number.
Step 2. $g(2) = 2^2 + 2 = 6$, and that whole value becomes the input of f: $f(6) = 3(6) - 1$.
So $(f \circ g)(2) = \mathbf{17}$.

Try it: With the same f and g, find $(g \circ f)(2)$. Does it match the example?

check: **27**

3. Invert a function

Write $y = f(x)$, swap x and y, solve for y. That is f^{-1}. Check it by composing: $f\big(f^{-1}(x)\big) = x$. Domain and range trade places, and the graph of f^{-1} is the graph of f reflected across $y = x$. An inverse exists only when no output repeats — otherwise restrict the domain until none does.

Example: $f(x) = 4x + 5$. Find f^{-1} and use it to compute $f^{-1}(13)$.
Step 1. An inverse undoes the operations in reverse order, and swapping the variables is the mechanical way to arrange that.
Step 2. $y = 4x+5$ swaps to $x = 4y+5$; subtracting 5 and dividing by 4 gives $f^{-1}(x) = \dfrac{x-5}{4}$, so $f^{-1}(13) = \dfrac{13-5}{4}$.

The value is **2**, and indeed $f(2) = 13$.

Try it: $f(x) = \dfrac{x}{2} - 6$. Find f^{-1} and compute $f^{-1}(-4)$.

check: **4**

(!) Watch out: f^{-1} is not $\dfrac{1}{f}$ — the -1 marks an inverse function, not a reciprocal. And undoing a composition reverses the order: $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$, socks on before shoes, shoes off before socks.